

Predicate Calculus

(First-Order Logic)

Syntax

A *first-order vocabulary* (or just *vocabulary* or *language*) $\mathcal{L}$ is specified by the following:

- 1) For each $n \in \mathbb{N}$ a set of n -ary function symbols (possibly empty). We use $f, g, h, \dots$ and also $+, \cdot, s$ as metasymbols for function symbols. A zero-ary function symbol is called a constant symbol.
- 2) For each $n \geq 0$, a set of n -ary predicate symbols (must be non-empty for some n). We use $P, Q, R, \dots$ and also $<, \leq, =$ as metasymbols for predicate symbols. A zero-ary predicate symbol is the same as a propositional atom.

In addition, the following symbols are available to build first-order formulas:

- 1) An infinite set of variables. We use $x, y, z, \dots$ and sometimes $a, b, c, \dots$ as metasymbols for variables. (Generally distinct letters x, y, z stand for distinct variables.)
- 2) connectives $\neg, \wedge, \vee$ (not, and, or)
- 3) quantifiers $\forall, \exists$ (for all, there exists)
- 4) $(,)$ (parentheses)

Terms and Formulas are built from these together with the function and predicate symbols from $\mathcal{L}$, as described below.

The standard vocabulary of arithmetic is

$$\mathcal{L}_A = [0, s, +, \cdot, =]$$

- 0 constant (zero-ary function symbol)
- s unary function symbol
- $+, \cdot$ binary function symbols
- $=$ binary predicate symbol

Terms (or *expressions*) are certain strings built from variables and function symbols, and are intended to represent objects in the universe of discourse.

Definition of an $\mathcal{L}$ -term (Here $\mathcal{L}$ is a first-order vocabulary):

- 1) Every variable is a term.
- 2) If f is an n -ary function symbol of $\mathcal{L}$ and $t_1, \dots, t_n$ are $\mathcal{L}$ -terms then $ft_1 \dots t_n$ is an $\mathcal{L}$ -term.

We will drop mention of $\mathcal{L}$ when it is not important, or clear from context.

Recall that a 0-ary function symbol is called a constant symbol (or sometimes just a *constant*). We use e as a metasyymbol for constants. Also 0 and 1 are constants. Note that all constants in $\mathcal{L}$ are $\mathcal{L}$ -terms.

Examples of $\mathcal{L}$ -terms (where f is binary and g is unary):
 $fgex, fxy, gfege$. These are parsed $f(g(e), x), f(x, y), g(f(e, g(e)))$ respectively.

Unique Readability Theorem for Terms: If terms $ft_1 \dots t_k$ and $fu_1 \dots u_\ell$ are syntactically equal, then $k = \ell$ and $t_i =_{syn} u_i$, $1 \leq i \leq k$.

Proof: Similar to the Unique Readability Theorem for propositional formulas (see page 2). To prove the lemma on weights, we assign a weight of $n - 1$ to each n -ary function symbol, and -1 to each variable. $\square$

Exercise 1 Carry out the details in the above argument.

Notation: We use $r, s, t, \dots$ to denote terms.

In the vocabulary for arithmetic $\mathcal{L}_A$, in practice we write $+, \cdot$ as though they were infix operators, even though officially they are prefix operators. Thus

Notation $(t_1 \cdot t_2) =_{syn} \cdot t_1 t_2$
 $(t_1 + t_2) =_{syn} + t_1 t_2$

Thus examples of our way of writing $\mathcal{L}_A$ terms are $sss0, ((x + sy) \cdot (ssz + s0))$

Definition of first-order formula in the vocabulary $\mathcal{L}$ (or $\mathcal{L}$ -formula, or just *formula*):

- 1) $Pt_1 \dots t_n$ is an *atomic* $\mathcal{L}$ -formula, where P is an n -ary predicate symbol in $\mathcal{L}$ and $t_1, \dots, t_n$ are $\mathcal{L}$ -terms.
- 2) If A and B are $\mathcal{L}$ -formulas, so are $\neg A$, $(A \wedge B)$, and $(A \vee B)$
- 3) If A is an $\mathcal{L}$ -formula and x is a variable, then $\forall x A$ and $\exists x A$ are $\mathcal{L}$ -formulas.

As in the case of propositional formulas, we use the notation

$(A \supset B)$ for $(\neg A \vee B)$

$(A \leftrightarrow B)$ for $(A \supset B) \wedge (B \supset A)$

Examples of formulas: $(\neg \forall x Px \vee \exists x \neg Px)$ (Here P is a unary predicate symbol.)

$(\forall x \neg Qxy \wedge \neg \forall z Qfyz)$. (Here Q is a binary predicate symbol and f is a unary function symbol.)

The Unique Readability Theorem holds for first-order formulas.

Notation $r = s$ stands for $= rs$

$r \neq s$ stands for $\neg(r = s)$

Example: Goldbach's conjecture: Every even integer greater than 2 is the sum of two primes.

$\forall x((\text{Even}(x) \wedge x > 2) \supset \exists y \exists z (\text{Prime}(y) \wedge \text{Prime}(z) \wedge x = y + z))$

Here Even, Prime are unary predicate symbols.

$>$ is a binary predicate symbol (we use infix notation).

2 is a constant symbol.

$+$ is a binary function symbol.

This can also be stated as a formula in the vocabulary $\mathcal{L}_A$, since the predicates Even, Prime, and $>$ can be defined in terms of $s, +, \cdot$, and $=$. For example, $\text{Even}(x)$ can be defined by the formula $\exists y(x = y + y)$.

Free and Bound Variables

Definition: An occurrence of x in A is *bound* iff it is in a subformula of A of the form $\forall x B$ or $\exists x B$. Otherwise the occurrence is *free*.

For example, in the formula $\exists y(x = y + y)$ (which defines $\text{Even}(x)$ as above) the occurrence of x is free, while the occurrences of y are bound. Intuitively the meaning of a formula depends on the values assigned to its free variables, but no value need be assigned to a bound variable to give the formula meaning.

Notice that a variable can have both free and bound occurrences in one formula. For example, in $Px \wedge \forall x Qx$, the first occurrence of x is free, and the second occurrence is bound.

Definition: A formula A or a term t is *closed* if it contains no free occurrence of a variable. A closed formula is called a *sentence*.

Semantics of Predicate Calculus

In the propositional calculus, a truth assignment provides meaning to a formula. In the predicate calculus, we need a more complicated object, called a *structure* (or *interpretation*)

to give meaning to formulas and terms. If $\mathcal{L}$ is a first-order vocabulary, then an $\mathcal{L}$ -structure $\mathcal{M}$ consists of the following:

- 1) A nonempty set M called the *universe of discourse* (or just *universe*). Variables in an $\mathcal{L}$ -formula range over M .
- 2) For each n -ary function symbol f in $\mathcal{L}$, an associated function $f^{\mathcal{M}} : M^n \mapsto M$. (If $n = 0$, then f is a constant symbol, and $f^{\mathcal{M}}$ is simply an element of M .)
- 3) For each n -ary predicate symbol in $\mathcal{L}$, an associated relation $P^{\mathcal{M}} \subseteq M^n$. If $\mathcal{L}$ contains $=$, then $=^{\mathcal{M}}$ must be the true equality relation on M .

Notice that the predicate symbol $=$ gets special treatment in the above definition, in that $=^{\mathcal{M}}$ must always be the true equality relation. Other predicate symbols may be interpreted by arbitrary relations of the appropriate arity. For example, if $\mathcal{L}$ contains the binary predicate symbol $<$, then $<^{\mathcal{M}}$ can be any binary relation on the universe M , and is not necessarily an order relation.

Every $\mathcal{L}$ -sentence becomes either true or false when interpreted by an $\mathcal{L}$ -structure $\mathcal{M}$, as explained below. If a sentence A becomes true under $\mathcal{M}$, then we say $\mathcal{M}$ *satisfies* A , or $\mathcal{M}$ is a *model* for A , and write $\mathcal{M} \models A$.

Definition: We say that a structure $\mathcal{M}$ is *finite* if the universe M of $\mathcal{M}$ is finite. Otherwise $\mathcal{M}$ is infinite.

If A has free variables, then these variables must be interpreted as specific elements in the universe M before A gets a truth value under the structure $\mathcal{M}$. For this we need the following:

Definition: An *object assignment* σ for a structure $\mathcal{M}$ is a mapping from variables to the universe M .

Below we give the formal definition of notion $\mathcal{M} \models \mathcal{A}[\sigma]$, which is intended to mean that the structure $\mathcal{M}$ satisfies the formula A when the free variables of A are interpreted according to the object assignment σ . First it is necessary to define the notation $t^{\mathcal{M}}[\sigma]$, which is the element of universe M assigned to the term t by the structure $\mathcal{M}$ when the variables of t are interpreted according to σ .

Basic Semantic Definition

Let $\mathcal{L}$ be a vocabulary, let $\mathcal{M}$ be an $\mathcal{L}$ -structure, and let σ be an object assignment for $\mathcal{M}$.

Each $\mathcal{L}$ -term t is assigned an element $t^{\mathcal{M}}[\sigma]$ in M , defined by structural induction on terms t , as follows (refer to the definition of $\mathcal{L}$ -term, page 19):

- a) $x^{\mathcal{M}}[\sigma]$ is $\sigma(x)$, for each variable x
- b) $(ft_1 \cdots t_n)^{\mathcal{M}}[\sigma] = f^{\mathcal{M}}(t_1^{\mathcal{M}}[\sigma], \dots, t_n^{\mathcal{M}}[\sigma])$

Notation: If x is a variable and $m \in M$, then the object assignment $\sigma(m/x)$ is the same as σ except $\sigma(m/x)(x) = m$.

For A an $\mathcal{L}$ -formula, the notion $\mathcal{M} \models A[\sigma]$ ($\mathcal{M}$ satisfies A under σ) is defined by structural induction on formulas A as follows (refer to the definition of formula):

- a) $\mathcal{M} \models (Pt_1 \cdots t_n)[\sigma]$ iff $\langle t_1^{\mathcal{M}}[\sigma], \dots, t_n^{\mathcal{M}}[\sigma] \rangle \in P^{\mathcal{M}}$
- b) $\mathcal{M} \models (s = t)[\sigma]$ iff $s^{\mathcal{M}}[\sigma] = t^{\mathcal{M}}[\sigma]$
- c) $\mathcal{M} \models \neg A[\sigma]$ iff not $\mathcal{M} \models A[\sigma]$.
- d) $\mathcal{M} \models (A \vee B)[\sigma]$ iff $\mathcal{M} \models A[\sigma]$ or $\mathcal{M} \models B[\sigma]$.
- e) $\mathcal{M} \models (A \wedge B)[\sigma]$ iff $\mathcal{M} \models A[\sigma]$ and $\mathcal{M} \models B[\sigma]$.
- f) $\mathcal{M} \models (\forall x A)[\sigma]$ iff $\mathcal{M} \models A[\sigma(m/x)]$ for all $m \in M$
- g) $\mathcal{M} \models (\exists x A)[\sigma]$ iff $\mathcal{M} \models A[\sigma(m/x)]$ for some $m \in M$

This method of giving meaning is sometimes called Tarski semantics, named after the important logician Alfred Tarski.

Note that item b) in the definition of $\mathcal{M} \models A[\sigma]$ follows from a) and the fact that $=^{\mathcal{M}}$ is always the equality relation.

If t is a closed term (i.e. contains no variables), then $t^{\mathcal{M}}[\sigma]$ is independent of σ , and so we sometimes just write $t^{\mathcal{M}}$. Similarly, if A is a sentence, then we sometimes write $\mathcal{M} \models A$ instead of $\mathcal{M} \models A[\sigma]$, since σ does not matter. (See the Corollary on the next page.)

Example: Let $\mathcal{L}$ be the vocabulary $\{; R, =\}$ and let $\mathcal{M}$ be the $\mathcal{L}$ -structure whose universe $M = \mathbb{N}$ and such that $R^{\mathcal{M}}(m, n)$ holds iff $m \leq n$. Then $\mathcal{M} \models \exists x \forall y R(x, y)$ (since 0 is the least element of $\mathbb{N}$) but $\mathcal{M} \not\models \exists y \forall x R(x, y)$ since there is no largest natural number.

Standard Structure: The *standard structure* $\underline{\mathbb{N}}$ for the vocabulary $\mathcal{L}_A$ has universe $M = \mathbb{N} = \{0, 1, 2, \dots\}$, $s^{\underline{\mathbb{N}}}(n) = n + 1$, and $0, +, \cdot, =$ get their usual meanings on the natural numbers.

Example: $\mathbb{N} \models \forall x \forall y \exists z (x + z = y \vee y + z = x)$ (since either $y - x$ or $x - y$ exists) but $\mathbb{N} \not\models \forall x \exists y (y + y = x)$ since not all natural numbers are even.

In the future we sometimes assume that there is some first-order vocabulary $\mathcal{L}$ in the background, and do not necessarily mention it explicitly.

Notation: In general, Φ denotes a set of formulas, $A, B, C, \dots$ denote formulas, $\mathcal{M}$ denotes a structure, and σ denotes an object assignment.

Lemma: If σ and σ' agree on the free variables of A , then $\mathcal{M} \models A[\sigma]$ iff $\mathcal{M} \models A[\sigma']$.

Proof: Structural induction on formulas A .

Corollary: If A is a sentence, then for any object assignments σ, σ' , $\mathcal{M} \models A[\sigma]$ iff $\mathcal{M} \models A[\sigma']$.

In view of the Corollary, if A is a sentence, then σ is irrelevant, so we omit mention of σ and simply write $\mathcal{M} \models A$.

Definition:

- a) A is *satisfiable* iff $\mathcal{M} \models A[\sigma]$ for some $\mathcal{M}$ and σ .
- b) $\mathcal{M} \models \Phi[\sigma]$ iff $\mathcal{M} \models A[\sigma]$ for all $A \in \Phi$. (We may omit mention of σ if Φ is a set of sentences.) We say Φ is *satisfiable* if $\mathcal{M} \models \Phi[\sigma]$ for some $\mathcal{M}$ and σ .
- c) $\Phi \models A$ iff for all $\mathcal{M}$ and all σ , if $\mathcal{M} \models \Phi[\sigma]$ then $\mathcal{M} \models A[\sigma]$.
- d) $\models A$ (A is *valid*) iff $\mathcal{M} \models A[\sigma]$ for all $\mathcal{M}$ and σ .
- e) $A \iff B$ (A and B are *logically equivalent*, or just *equivalent*) iff for all $\mathcal{M}$ and all σ , $\mathcal{M} \models A[\sigma]$ iff $\mathcal{M} \models B[\sigma]$.

$\Phi \models A$ is read “ A is a logical consequence of Φ ”. This relation is of FUNDAMENTAL IMPORTANCE. Do not confuse this with our other use of the symbol $\models$, as in $\mathcal{M} \models A$ ($\mathcal{M}$ satisfies A). In the latter, $\mathcal{M}$ is a structure, rather than a set of formulas.

Note that $\models$ is a symbol of the “meta language” (English), as opposed to $\neg, \vee, \wedge, \forall, \exists$, which are symbols of the “object language”.

As in the propositional case, if $\Phi = \{B_1, \dots, B_n\}$, then we sometimes write $B_1, \dots, B_n \models A$ instead of $\{B_1, \dots, B_n\} \models A$.

Examples:

- 1 $(\forall x A \vee \forall x B) \models \forall x (A \vee B)$, for all formulas A and B .

Proof: We follow the definition of $\Phi \models A$ above. Let $\mathcal{M}$ be any structure and let σ be any object assignment. Assume L.H.S. is true, i.e. $\mathcal{M} \models (\forall x A \vee \forall x B)[\sigma]$.

Then following the Basic Semantic Definition, $\mathcal{M} \models (\forall xA)[\sigma]$ or $\mathcal{M} \models (\forall xB)[\sigma]$. Say $\mathcal{M} \models (\forall xA)[\sigma]$. Then $\mathcal{M} \models A[\sigma(m/x)]$ for all $m \in M$. Then $\mathcal{M} \models (A \vee B)[\sigma(m/x)]$ for all $m \in M$. Therefore $\mathcal{M} \models \forall x(A \vee B)[\sigma]$.

Similarly for the case $\mathcal{M} \models (\forall xB)[\sigma]$. $\square$

2 $\forall x(A \vee B) \models (\forall xA \vee \forall xB)$? No, not necessarily.

Take $A =_{syn} Px$, $B =_{syn} Qx$, define the structure $\mathcal{M}$ to have universe $M = \mathbb{N}$, define $P^{\mathcal{M}}$ to be the set of even natural numbers, and $Q^{\mathcal{M}}$ to be the set of odd natural numbers. Then $\mathcal{M} \models \forall x(Px \vee Qx)$ (every number is even or odd), but not $\mathcal{M} \models (\forall xPx \vee \forall xQx)$ (it is not the case that either all numbers are even or all numbers are odd).

3 $\neg \forall xA \iff \exists x \neg A$, for all formulas A .

$\neg \exists xA \iff \forall x \neg A$, for all formulas A .

$(\forall xA \wedge \forall xB) \iff \forall x(A \wedge B)$, for all formulas A, B .

$\exists x(A \vee B) \iff (\exists xA \vee \exists xB)$, for all formulas A, B .

$\exists x(A \wedge B) \models (\exists xA \wedge \exists xB)$, for all formulas A, B .

NOT $(\exists xA \wedge \exists xB) \models \exists x(A \wedge B)$ in general

$\forall x \forall y A \iff \forall y \forall x A$

$\exists x \exists y A \iff \exists y \exists x A$

$\exists y \forall x A \models \forall x \exists y A$, for all formulas A .

NOT $\forall x \exists y A \models \exists y \forall x A$ in general

$\forall x A \models \exists x A$, because of our requirement that every universe M must be nonempty.

$\forall x \forall y (x = y \supset fx = fy)$ is valid.

$\forall x \forall y (fx = fy \supset x = y)$ is NOT valid.

Exercise 2 Verify each line in item 3 above. For the two lines beginning NOT give specific formulas A (and B) for which the relation is false, and show it is false by giving a specific structure which satisfies the left hand side but not the right hand side. For the last line, give a structure which does not satisfy the formula.

Exercise 3 Show that $\{P0, Ps0, Pss0, \dots\} \not\models \forall x Px$ by giving a specific structure.

Exercise 4 Consider the following four formulas over the vocabulary $\mathcal{L}_A$:

$P1: \forall x (sx \neq 0)$

$P2: \forall x \forall y (sx = sy \supset x = y)$

$P3: \forall x (x + 0 = x)$

$P4: \forall x \forall y (x + sy = s(x + y))$

Prove from the definition of $\models$ that

$$P1, P2, P3, P4 \not\models \forall x \forall y (x + y = y + x)$$

Hint: Think of $+$ as string concatenation.

Exercise 5 Show that $\forall x (gfx = x)$ is NOT a logical consequence of $\forall x (fgx = x)$.

Exercise 6 Let $\mathcal{M}$ be a structure and let Φ be the set of all sentences A satisfied by $\mathcal{M}$. Show that Φ is closed under $\models$. That is, show that if $\Phi \models A$ then $A \in \Phi$.

Exercise 7 Give a sentence in the vocabulary $\mathcal{L} = \{; =\}$ which is satisfied by a structure iff the universe has exactly three elements.

Exercise 8 Give a satisfiable sentence A in the vocabulary $\mathcal{L} = \{; R\}$, where R is a binary predicate symbol, such that A has no finite model. (Hint: Think of R as an order relation.)

Exercise 9 Give a sentence A in the vocabulary $\mathcal{L} = \{; R, =\}$, where R is a binary predicate symbol, such that for all $n \in \mathbb{N}, n > 0$, A has a model whose universe has n elements iff n is even. (Hint: Think of R as a pairing relation.)

Exercise 10 Give a sentence A of the predicate calculus with the vocabulary $\mathcal{L} = \{; R, =\}$, where R is a binary predicate symbol, such that a finite $\mathcal{L}$ -structure (thought of as a directed graph with edge relation R) is a model for A iff it is a union of disjoint directed cycles. Now give an infinite model for A .

Recall that a sentence is a formula with no free variables. Each sentence in the vocabulary $\mathcal{L}_{\mathcal{A}}$ (the vocabulary of arithmetic) is either true or false in the standard structure $\mathbb{N}$. Thus $\forall x \forall y (x + y = y + x)$ and Fermat's Last Theorem are true, while $\forall x \neg (0 = x + x)$ is false, and no one knows the truth value of Goldbach's conjecture. On the other hand, a formula such as $\forall y \neg (x = y + y)$ ("x is odd") has no truth value under any structure, since it has a free variable. Of course it gets a truth value in a structure when an object assignment σ is specified.

Substitution

Syntactic Definition: (t, u are terms)

$t(u/x)$ is the result of replacing all occurrences of x in t by u .

$A(u/x)$ is the result of replacing all *free* occurrences of x in A by u .

Semantics:

Lemma For each structure $\mathcal{M}$ and each object assignment σ ,

$$(t(u/x))^{\mathcal{M}}[\sigma] = t^{\mathcal{M}}[\sigma(m/x)]$$

where $m = u^{\mathcal{M}}[\sigma]$.

Example: Let $\mathcal{M}$ be the standard structure $\mathbb{N}$ for the vocabulary $\mathcal{L}_A$ of arithmetic. Suppose $\sigma(x) = 5$ and $\sigma(y) = 7$. Let t be the term $x + y$ and let u be the term $ss0$ (here s is the successor function in $\mathcal{L}$). Then $t(u/x)$ is $ss0 + y$ and so $(t(u/x))^{\mathbb{N}}[\sigma] = 2 + 7 = 9$. On the other hand, $m = u^{\mathbb{N}} = 2$, so $t^{\mathbb{N}}[\sigma(m/x)] = 2 + 7 = 9$, and the Lemma is verified for this case.

Proof if the Lemma: Structural induction on t .

Base case: t is a variable. If the variable is x , then both sides of the equation are the same, namely $u^{\mathcal{M}}[\sigma]$. If t is a variable y other than x , then again both sides are the same, namely $\sigma(y)$.

The induction step is straightforward from the Basic Semantic Definition. $\square$

Exercise 11 Carry out the induction step in detail.

Question: Does the above lemma apply to formulas A ? I.e. can we say $\mathcal{M} \models A(t/x)[\sigma]$ iff $\mathcal{M} \models A[\sigma(m/x)]$, where $m = t^{\mathcal{M}}[\sigma]$? Something can go wrong.

Example: Suppose A is $\forall y \neg(x = y + y)$. This says “ x is odd”. But $A(x + y/x)$ is $\forall y \neg(x + y = y + y)$, which does not say “ $x + y$ is odd” as desired, but instead it is always false. The problem is that y in the term $x + y$ got “caught” by the quantifier $\forall y$.

Definition A term t is *freely substitutable* for x in A iff no free occurrence of x in A is in a subformula of A of the form $\forall y B$ or $\exists y B$, where y occurs in t .

Substitution Theorem: If t is freely substitutable for x in A then for all structures $\mathcal{M}$ and all object assignments σ , $\mathcal{M} \models A(t/x)[\sigma]$ iff $\mathcal{M} \models A[\sigma(m/x)]$, where $m = t^{\mathcal{M}}[\sigma]$.

Proof: Structural induction on A . The interesting case is when A is $\forall y B$. (The case when A is $\exists y B$ is similar). Then we are to prove

$$\mathcal{M} \models (\forall y B)(t/x)[\sigma] \text{ iff } \mathcal{M} \models (\forall y B)[\sigma(m/x)] \quad (1)$$

where $m = t^{\mathcal{M}}[\sigma]$.

If x does not occur free in $\forall y B$, then no substitution is done, so the result is easy. (If x, y are the same variable, then x does not occur free in $\forall y B$.)

Hence we may assume that x, y are distinct variables and x occurs free in B . Since t is freely substitutable for x in $\forall y B$, y does not occur in t .

Following the Basic Semantic Definition, the LHS of (1) holds iff $\mathcal{M} \models B(t/x)[\sigma(n/y)]$ for all $n \in M$. Apply the induction hypothesis to B to obtain

$$\mathcal{M} \models B(t/x)[\sigma(n/y)] \text{ iff } \mathcal{M} \models B[\sigma(n/y)(m'/x)]$$

where now $m' = t^{\mathcal{M}}[\sigma(n/y)]$. But note that $m' = t^{\mathcal{M}}[\sigma(n/y)] = t^{\mathcal{M}}[\sigma] = m$ because y does not occur in t . Hence

$$\mathcal{M} \models B(t/x)[\sigma(n/y)] \text{ iff } \mathcal{M} \models B[\sigma(n/y)(m/x)]$$

Now the RHS of (1) holds iff $\mathcal{M} \models B[\sigma(m/x)(n/y)]$ for all $n \in M$. But $\sigma(n/y)(m/x) = \sigma(m/x)(n/y)$, since x and y are distinct. Hence the LHS holds iff the RHS holds. $\square$

Change of Bound Variable

If a term t is not freely substitutable for x in A , it is because some variable y in t gets caught by a quantifier $\forall y$ or $\exists y$ in A . One way to fix this is simply rename the bound variable y in A to some new variable z . It should be intuitively clear that this renaming does not change the meaning of A . The definition and lemmas below formalize this process.

Definition: $\forall zA(z/y)$ results from $\forall yA$ by *change of bound variable* provided z does not occur in A . Similarly for $\exists zA(z/y)$.

Lemma: If z does not occur in A , then $\forall zA(z/y)$ and $\forall yA$ are logically equivalent. Also $\exists zA(z/y)$ and $\exists yA$ are equivalent.

Proof: This follows from the Basic Semantic Definition and the Substitution Theorem. (Verify this). $\square$

Definition A' is a *variant* of A if A' results by a sequence of changes of bound variables to subformulas of A .

Theorem: If A' is a variant of A then A and A' are equivalent.

This follows from the preceding Lemma and the following general result:

Replacement Theorem: If B and B' are equivalent formulas and A' results from A by replacing some occurrence of B in A by B' , then A and A' are equivalent.

Exercise 12 Prove the Replacement Theorem, by structural induction on A (relative to B). The base case is when A and B coincide.

Example: B is $\neg\forall xPxy$, B' is $\exists z\neg Pzy$, A is $\forall y(\neg\forall xPxy \supset Qy)$. Note that B has a free variable that is bound in A . A' is $\forall y(\exists z\neg Pzy \supset Qy)$. By the Replacement Theorem, A and A' are equivalent, even though the quantifier $\forall y$ in A catches a variable in B .

A First-Order Gentzen System

We now extend the propositional proof system PK to the first-order sequent proof system LK . For this it is convenient to introduce two kinds of variables:

- type “free”: $a, b, c, \dots$

- type “bound”: $x, y, z, \dots$

A first-order formula A is called a *proper formula* if it satisfies the restriction that every variable that occurs free has type free, and every variable that occurs bound has type bound. Similarly a *proper term* has no variable of type bound. Notice that a subformula of a proper formula is not necessarily proper, and a proper formula may contain terms which are not proper.

The sequent system LK is an extension of the propositional system PK , where now all formulas $A_1, \dots, A_k, B_1, \dots, B_\ell$ in a sequent $A_1, \dots, A_k \rightarrow B_1, \dots, B_\ell$ must be proper formulas. In addition to the rules given for PK , the system LK has four rules for introducing the quantifiers.

Notation: In the rules below, t is any proper term and $A(t)$ is the result of substituting t for all free occurrences of x in $A(x)$. Similarly $A(b)$ is the result of substituting b for all free occurrences of x in $A(x)$. Note that t and b can always be freely substituted for x in $A(x)$ because $\forall x A(x)$ and $\exists x A(x)$ are proper formulas.

$\forall$ introduction rules

$$\text{left} \frac{A(t), \Gamma \rightarrow \Delta}{\forall x A(x), \Gamma \rightarrow \Delta} \qquad \text{right} \frac{\Gamma \rightarrow \Delta, A(b)}{\Gamma \rightarrow \Delta, \forall x A(x)}$$

$\exists$ introduction rules

$$\text{left} \frac{A(b), \Gamma \rightarrow \Delta}{\exists x A(x), \Gamma \rightarrow \Delta} \qquad \text{right} \frac{\Gamma \rightarrow \Delta, A(t)}{\Gamma \rightarrow \Delta, \exists x A(x)}$$

Restriction: The free variable b must not occur in the conclusion in $\forall$ **right** and $\exists$ **left**.

Example: An instance of $\forall$ -**left** is

$$\frac{Pbb \rightarrow Pbb}{\forall y Pby \rightarrow Pbb}$$

What is the formula $A(y)$ in this case?

Semantics of first-order sequents

The semantics of first-order sequents is a natural generalization of the semantics of propositional sequents given on page 10. Again a sequent $S =_{syn}$

$$A_1, \dots, A_k \rightarrow B_1, \dots, B_\ell$$

has the same meaning as its associated formula $A_S =_{syn}$

$$(A_1 \wedge A_2 \wedge \dots \wedge A_k) \supset (B_1 \vee B_2 \vee \dots \vee B_\ell) \tag{2}$$

In particular, we say that the sequent is *valid* iff its associated formula is valid.

Definition: [Universal Closure] Suppose that A is a formula whose free variables comprise the list $a_1, \dots, a_n$. Then the *universal closure* of A , written $\forall A$, is the sentence $\forall x_1 \dots \forall x_n A(x_1/a_1, \dots, x_n/a_n)$, where $x_1, \dots, x_n$ are new (bound) variables. If Φ is a set of formulas, then $\forall \Phi$ is the set of all sentences $\forall A$, for A in Φ .

Note that every formula A is valid iff its universal closure $\forall A$ is valid. Also A is a logical consequence of its universal closure $\forall A$, but $\forall A$ is not necessarily a logical consequence of A (for example take $A =_{syn} Pa$).

Recall that for the propositional system PK , for each rule the bottom sequent is a logical consequence of the top sequent(s). This remains true for LK , with the exception of the rules $\forall$ -**right** and $\exists$ -**left**. For these rules we can make a weaker statement: the universal closure of (the meaning of) the bottom sequent is a logical consequence of the universal closure of (the meaning of) the top sequent. The following proposition makes this weaker statement for all the PK rules. (The statement is weaker, because for any formulas A and B , if $A \models B$, then $\forall A \models \forall B$).

Lemma For each LK rule, the universal closure of the meaning of the bottom sequent is a logical consequence of the universal closure(s) of the meaning(s) of the top sequent(s). Here the *meaning* of a sequent S is the formula A_S given in (2).

Proof: The argument for the propositional rules is essentially the same as for the system PK . The arguments for $\forall$ -**left** and $\exists$ -**right** are easy; and in fact in these cases it is not necessary to take universal closures.

We illustrate the remaining two rules by considering the case of $\forall$ -**right**. Note that because of the **Restriction** for this rule, the variable b cannot occur in Γ or Δ . Hence it suffices to verify that

$$\forall x(\bigwedge \Gamma \supset (\bigvee \Delta \vee A(x))) \models \bigwedge \Gamma \supset (\bigvee \Delta \vee \forall x A(x))$$

To see that this logical consequence holds, suppose that $\mathcal{M}$ is a structure and σ is an object assignment. Suppose that $\mathcal{M}$ satisfies the left hand side under σ , i.e.

$$\mathcal{M} \models \forall x(\bigwedge \Gamma \supset (\bigvee \Delta \vee A(x)))[\sigma]$$

Either $\mathcal{M}$ satisfies $\forall x A(x)$ under σ or not. In the first case it follows immediately that $\mathcal{M}$ satisfies the right hand side under σ . In the second case, it must be that

$$\mathcal{M} \models \forall x(\bigwedge \Gamma \supset \bigvee \Delta)[\sigma]$$

and hence again $\mathcal{M}$ satisfies the right hand side under σ . □

Exercise 13 Give the argument for the other three quantifier rules.

Soundness Theorem for LK : Every sequent provable in LK is valid.

Proof: This is proved by induction on the number of sequents in the LK proof. For the base case, obviously each axiom $A \rightarrow A$ is valid. For the induction step, it follows from the

above lemma that for each rule, if all sequents on top are valid, then the sequent on the bottom is valid. $\square$

Exercise 14 Give a specific example of a sequent $\Gamma \rightarrow \Delta, A(b)$ which is valid, but the bottom sequent $\Gamma \rightarrow \Delta, \forall x A(x)$ is not valid, because the restriction for the $\forall$ **right** rule is violated (i.e. b occurs in Γ or Δ or $\forall x A(x)$). Do the same for the $\exists$ **left** rule.

An *LK* proof of a valid first-order sequent can be obtained using the same method as in the propositional case: Write the goal sequent at the bottom, and move up by using the introduction rules in reverse. A good heuristic is: if there is a choice about which quantifier to remove next, choose $\forall$ **right** and $\exists$ **left** first (working backwards), since these rules carry a restriction.

Here is an *LK* proof of the sequent $(\forall x Px \vee \forall x Qx) \rightarrow \forall x (Px \vee Qx)$.

$$\begin{array}{c}
 \frac{Pb \rightarrow Pb}{Pb \rightarrow Pb, Qb} (\text{weakening}) \quad \frac{Qb \rightarrow Qb}{Qb \rightarrow Pb, Qb} (\text{weakening}) \\
 \frac{Pb \rightarrow Pb, Qb}{Pb \rightarrow (Pb \vee Qb)} (\vee \text{ right}) \quad \frac{Qb \rightarrow Pb, Qb}{Qb \rightarrow (Pb \vee Qb)} (\vee \text{ right}) \\
 \frac{Pb \rightarrow (Pb \vee Qb)}{\forall x Px \rightarrow (Pb \vee Qb)} (\forall \text{ left}) \quad \frac{Qb \rightarrow (Pb \vee Qb)}{\forall x Qx \rightarrow (Pb \vee Qb)} (\forall \text{ left}) \\
 \hline
 \frac{\forall x Px \rightarrow (Pb \vee Qb) \quad \forall x Qx \rightarrow (Pb \vee Qb)}{(\forall x Px \vee \forall x Qx) \rightarrow (Pb \vee Qb)} (\vee \text{ left}) \\
 \hline
 \frac{(\forall x Px \vee \forall x Qx) \rightarrow (Pb \vee Qb)}{(\forall x Px \vee \forall x Qx) \rightarrow \forall x (Px \vee Qx)} (\forall \text{ right})
 \end{array}$$

Exercise 15 Give *LK* proofs for the following valid sequents:

$$\begin{aligned}
 &\forall x Px \wedge \forall x Qx \rightarrow \forall x (Px \wedge Qx) \\
 &\forall x (Px \wedge Qx) \rightarrow \forall x Px \wedge \forall x Qx \\
 &\exists x (Px \vee Qx) \rightarrow \exists x Px \vee \exists x Qx \\
 &\exists x Px \vee \exists x Qx \rightarrow \exists x (Px \vee Qx) \\
 &\exists x (Px \wedge Qx) \rightarrow \exists x Px \wedge \exists x Qx \\
 &\exists y \forall x Pxy \rightarrow \forall x \exists y Pxy \\
 &\forall x Px \rightarrow \exists x Px
 \end{aligned}$$

Check that the rule restrictions seem to prevent generating *LK* proofs for the following invalid sequents:

$$\begin{aligned}
 &\exists x Px \wedge \exists x Qx \rightarrow \exists x (Px \wedge Qx) \\
 &\forall x \exists y Pxy \rightarrow \exists y \forall x Pxy
 \end{aligned}$$